Project Title: Serverless E-Commerce ETL Pipeline on AWS

Overview:

Built a fully automated, serverless ETL pipeline on AWS that ingests raw e-commerce sales data, transforms it using PySpark, and serves business insights through SQL queries and a live dashboard.

Architecture:

CSV Upload → S3 (Raw) → Lambda → Glue (PySpark) → S3 (Parquet) → Athena (SQL) → QuickSight (Dashboard)

Tech Stack:

Python, PySpark, SQL, AWS S3, AWS Glue, AWS Lambda, Amazon Athena, Amazon QuickSight, AWS IAM

What I Built:

- Generated a realistic e-commerce dataset using Python
- Stored raw CSV data in Amazon S3
- Automated pipeline trigger using AWS Lambda on file upload
- Cleaned and transformed data using AWS Glue (PySpark)
- Stored processed data as Parquet partitioned by region/year
- Queried data using Amazon Athena (SQL)
- Built a business dashboard in Amazon QuickSight

Key Transformations:

- Removed null and invalid records
- Cast dates and numeric columns to correct types
- Derived new columns: days_to_ship, profit_margin, shipping_speed, order_size, is_profitable
- Separated cancelled orders into dedicated folder

Dashboard Charts:

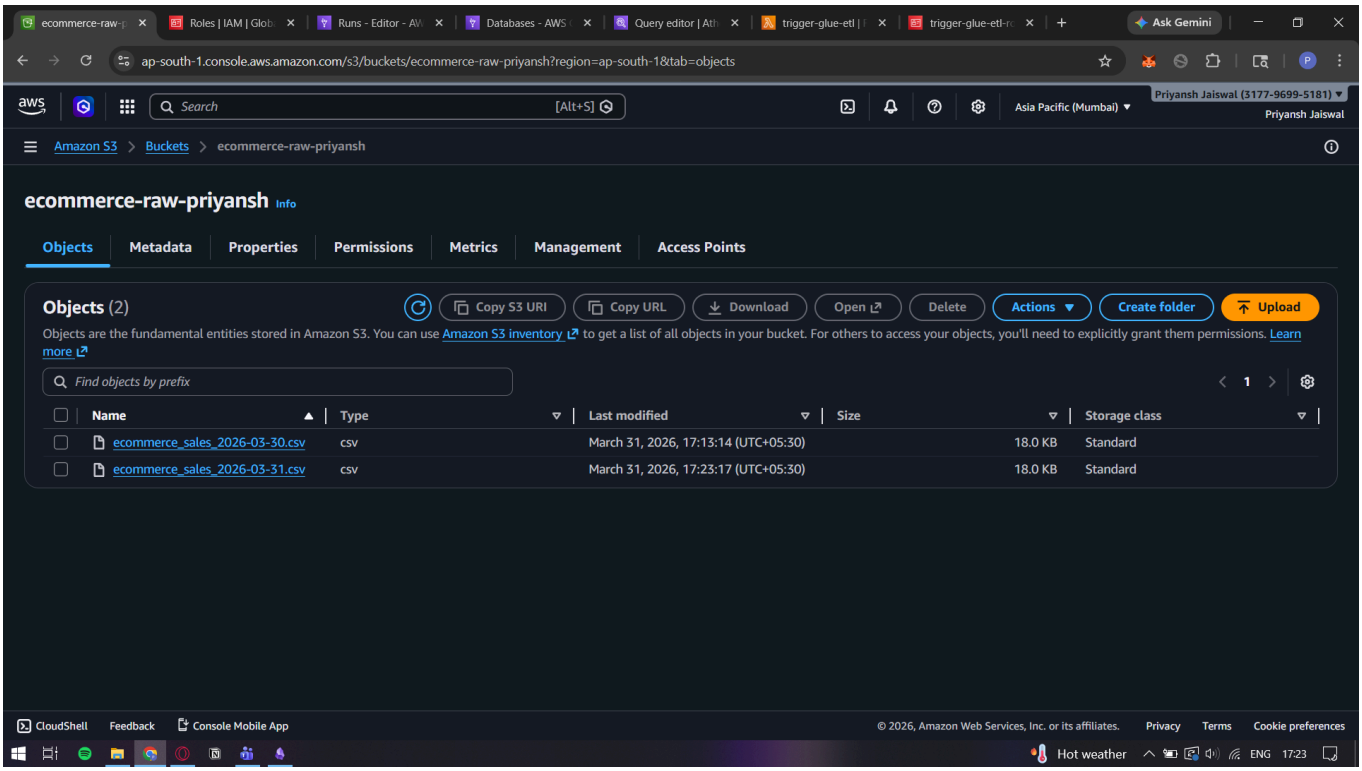
- Revenue by region (bar chart)

- Top products by revenue (horizontal bar)
- Orders by category (pie chart)
- Monthly revenue trend (line chart)

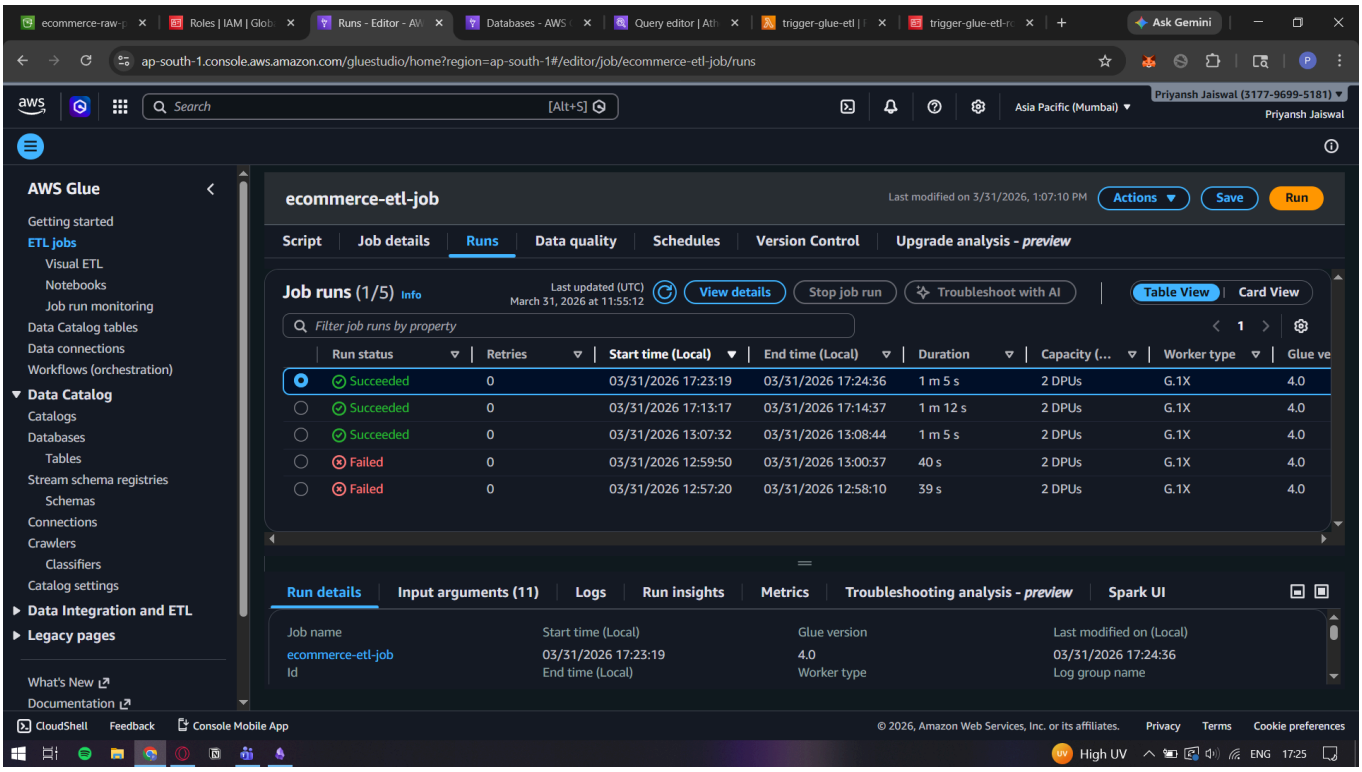
Outcome:

A fully automated, event-driven data pipeline deployed on AWS — ready to process any new sales file uploaded to S3 without any manual intervention.

1. Both S3 buckets with files



2. Glue job run history showing successful runs



3. Athena query results

Query 1 — Total revenue by region:

```
SELECT region,
COUNT(*) as total_orders,
ROUND(SUM(revenue), 2) as total_revenue,
ROUND(SUM(profit), 2) as total_profit
FROM sales
GROUP BY region
ORDER BY total_revenue DESC;
```

#	region	total_orders	total_revenue	total_profit
1	Central	20	93943.8	39863.65
2	West	16	48729.25	19332.3
3	South	14	46303.3	24464.15
4	East	15	44034.8	14791.7
5	North	13	42803.6	19507.3

Query 2 — Top 5 products by revenue:

```
SELECT product_name,
category,
COUNT(*) as orders,
ROUND(SUM(revenue), 2) as total_revenue
FROM sales
GROUP BY product_name, category
ORDER BY total_revenue DESC
LIMIT 5;
```

Query results | Query stats

Completed — Time in queue: 52 ms — Run time: 597 ms — Data scanned: 3.45 KB

Results (5) — Copy — Download results CSV

Search rows — 1

#	product_name	category	orders	total_revenue
1	Mechanical Keyboard	Electronics	8	57102.15
2	Running Shoes	Footwear	4	34025.2
3	Bluetooth Speaker	Electronics	6	30784.6
4	Laptop Stand	Electronics	6	29682.15
5	Backpack 30L	Bags	5	22934.7

Query 3 — Monthly sales trend:

```
SELECT order_year,
order_month,
COUNT(*) as total_orders,
ROUND(SUM(revenue), 2) as monthly_revenue
FROM sales
GROUP BY order_year, order_month
ORDER BY order_year, order_month;
```

Query results | Query stats

Completed — Time in queue: 91 ms — Run time: 606 ms — Data scanned: 1.46 KB

Results (6) — Copy — Download results CSV

Search rows

#	order_year	order_month	total_orders	monthly_revenue
1	2025	10	9	22503.9
2	2025	11	13	53556.5
3	2025	12	11	47504.45
4	2026	1	16	58347.35
5	2026	2	17	59649.25
6	2026	3	12	34253.3

Query 4 — Average shipping speed by region:

```
SELECT region,
ROUND(AVG(days_to_ship), 1) as avg_days_to_ship,
COUNT(*) as total_orders
FROM sales
GROUP BY region
ORDER BY avg_days_to_ship;
```

Query results | Query stats

Completed — Time in queue: 135 ms — Run time: 767 ms — Data scanned: 0.69 KB

Results (5) — Copy — Download results CSV

Search rows

#	region	avg_days_to_ship	total_orders
1	Central	3.8	20
2	West	4.0	16
3	North	4.1	13
4	South	4.3	14
5	East	4.4	15

4. QuickSight dashboard with all 4 charts

Chart 1 — Revenue by Region (already set up, just add the value)

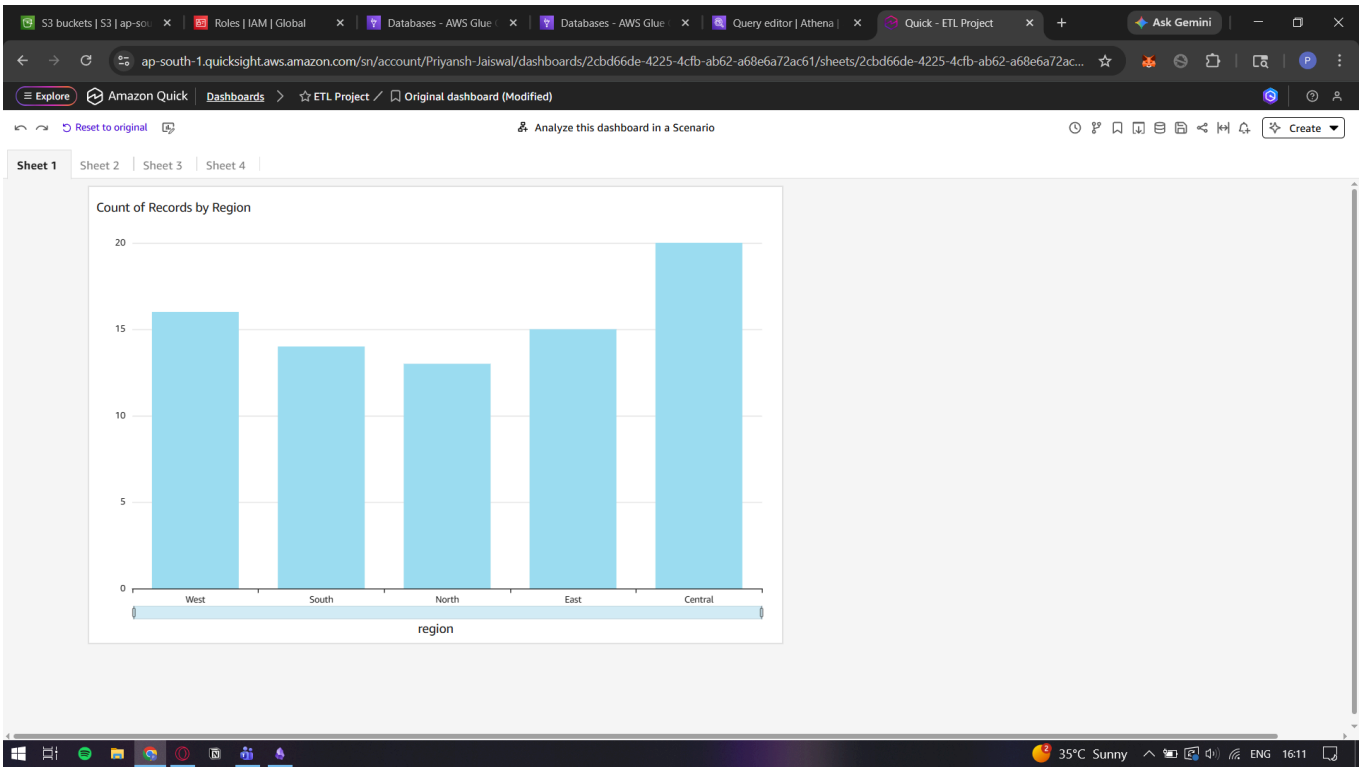


Chart 2 — Top Products by Revenue

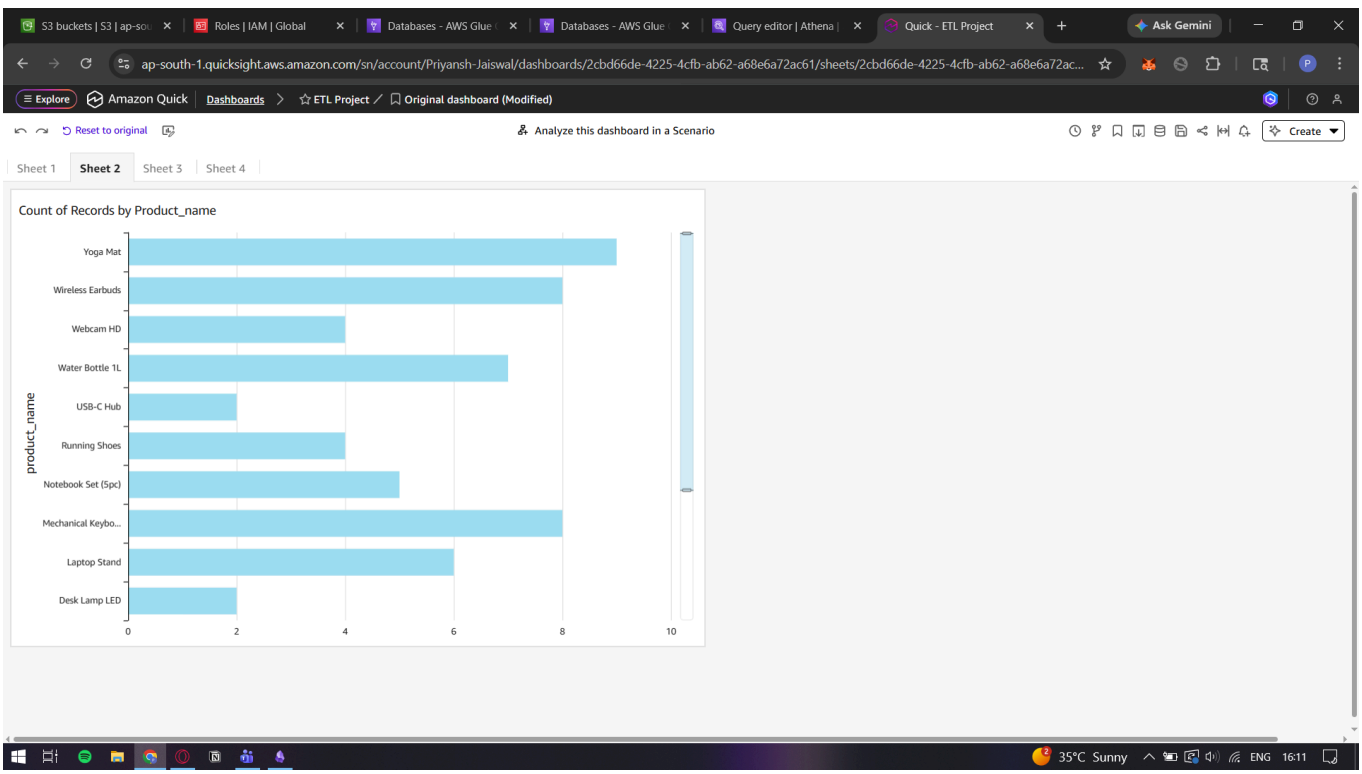


Chart 3 — Orders by Category (Pie Chart)

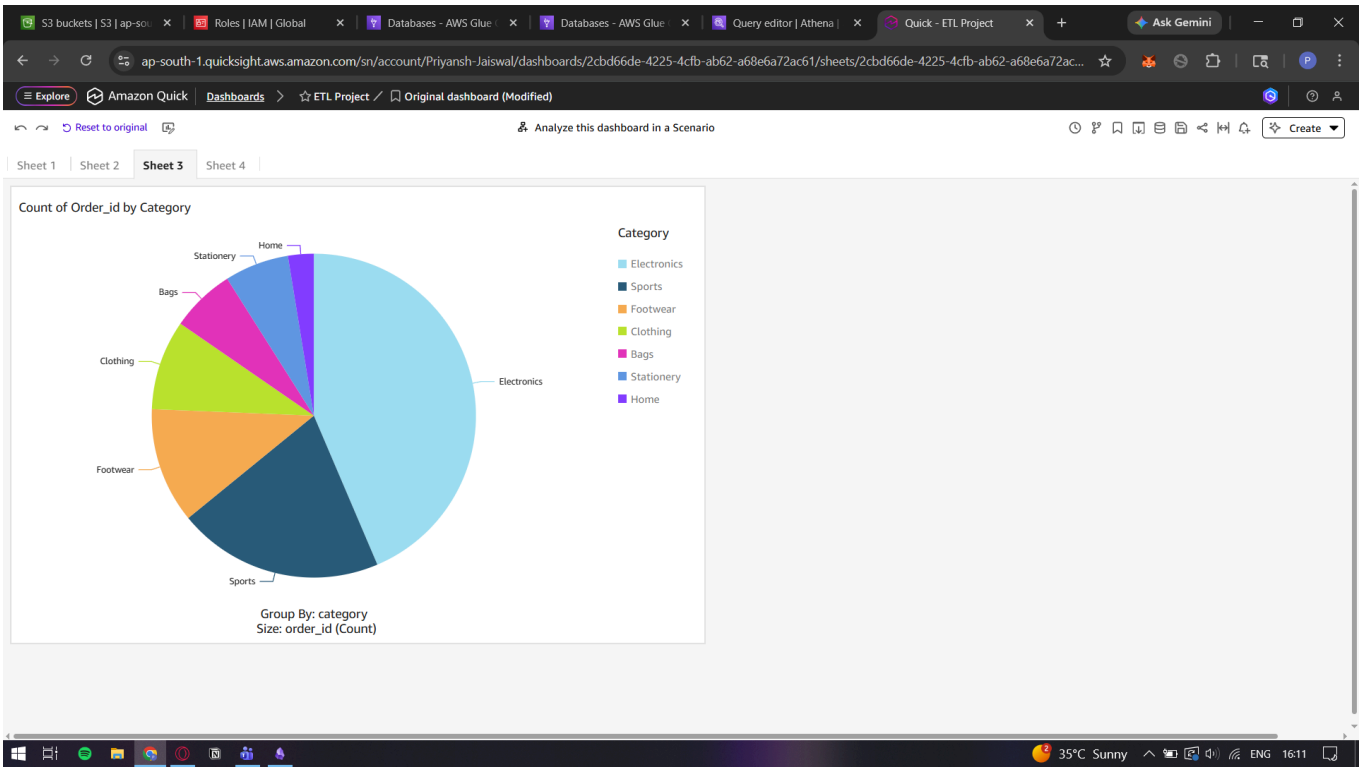
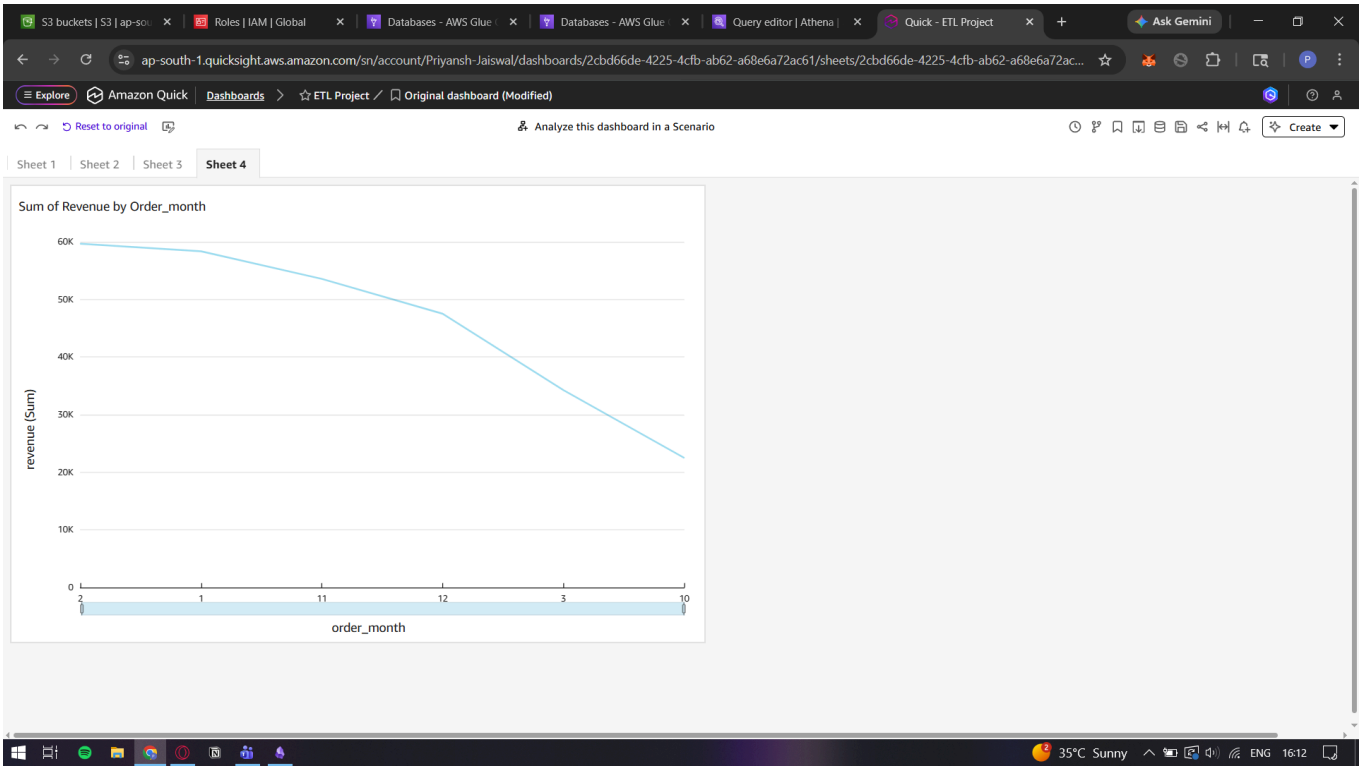



Chart 4 — Monthly Revenue Trend